

THE COOKIE LAB

Retail Analytics & Business Intelligence

SQL Server → Business Analysis → Power BI

PORTFOLIO CASE STUDY

A decision-oriented retail analytics case study covering revenue, customers, RFM segmentation, products, stores, refunds, churn risk and retention.

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Core Tools: Microsoft SQL Server (SSMS) | Power BI

Table of Contents

1. Executive Summary.....	3
2. Business Problem & Analytical Objectives.....	4
3. Company Context & Data Architecture.....	4
4. Data Dictionary — Analytical Fields.....	5
5. Data Preparation & Analytical Controls.....	6
6. Analytical Question Framework.....	7
7. SQL Server Analysis.....	7
7.1 Revenue, Channel & Transaction Performance.....	7
7.2 Monthly Sales Trend & Growth.....	8
7.3 Customer RFM Segmentation.....	8
7.4 Product & Category Performance.....	9
7.5 Store Performance & Customer Lifetime Value.....	10
7.6 Refund Hotspots & Churn Risk.....	10
7.7 Cohort & Customer Retention.....	11
8. From SQL Analysis to Power BI.....	12
9. Power BI Dashboard Overview.....	12
10. Power BI Page 1 — Retail Sales, Customer & Operational Performance.....	13
11. Power BI Page 2 — Sales & Channel Performance.....	14
12. Power BI Page 3 — Customer Intelligence & RFM Analysis.....	15
13. Power BI Page 4 — Product & Store Performance.....	15
14. Cross-Dashboard Business Insights.....	16
15. Business Recommendations.....	17
16. Final Conclusion.....	18
17. Analyst Perspective.....	19
18. Technical Skills & Tools.....	19
19. Portfolio Project Summary.....	20

1. Executive Summary

The Cookie Lab Retail Analytics project was built to answer a practical business question: how can a growing omnichannel cookie retailer use its operational and customer data to understand performance, identify risk, and make better commercial decisions?

The project follows a complete analyst workflow. A relational retail dataset containing more than 824,838 transaction records was analyzed in SQL Server through a series of business-focused analytical questions. The resulting findings were then translated into a four-page Power BI reporting solution built for executive review and deeper operational analysis.

THE ANALYTICAL STORY

SQL was used to discover what is happening in the business; Power BI was used to communicate, explore, and monitor those insights.

Project Overview

Area	Portfolio Output
Business Context	Canadian premium omnichannel cookie retailer
Data Scale	824,838+ transaction records plus interconnected operational tables
SQL Analysis	Revenue, trends, RFM, products, stores, LTV, refunds, churn, and retention
BI Layer	Four-page Power BI dashboard
Decision Focus	Growth, customer value, operational risk, product performance, and store performance

Headline Findings

- Gross Revenue: 82.60M; Net Revenue: 78.94M.
- Total Refunds: 3.66M, creating a clear operational monitoring opportunity.
- Total Orders: 750K and Total Customers: 115K.
- In-Store is the leading sales channel, followed by Mobile App and Website.
- Credit Card is the dominant payment method, generating approximately 36M revenue.
- Cookie Cake (3) is the leading product by gross revenue at approximately 7.1M.
- Champions/VIP customers show the strongest monetary value and purchase frequency.
- Holiday Cookie Tin leads unit sales at approximately 233K units and also carries the highest refund amount at approximately 0.76M.
- The Cookie Lab London Main Street is the top-performing store at approximately 4.8M gross revenue.

2. Business Problem & Analytical Objectives

Business Problem

The Cookie Lab operates across physical stores and digital channels, generating a large volume of transaction, customer, product, and operational data. The business needs a consolidated view of commercial performance and customer behavior rather than a set of isolated reports.

The analytical challenge was to move from raw relational data to a decision-ready view that could answer where revenue is coming from, which customers matter most, which products and stores lead performance, where refunds create risk, and which customers may require retention action.

Analytical Objectives

- Measure revenue, net sales, refunds, and transaction performance by channel and payment method.
- Identify monthly sales patterns and recurring revenue seasonality.
- Segment customers using Recency, Frequency, and Monetary (RFM) behavior.
- Identify leading products and categories by order frequency, units sold, and gross revenue.
- Compare store-level revenue performance and identify leading locations.
- Identify high-value customers using lifetime spending and order activity.
- Detect high-value customers with elevated churn risk based on purchase recency.
- Identify product-level refund hotspots and potential quality or fulfillment issues.
- Evaluate customer retention using monthly cohort analysis.

DECISION PRINCIPLE

Every analytical question was selected to connect a measurable business outcome with a potential management action.

3. Company Context & Data Architecture

Company Overview

The Cookie Lab is positioned as a premium Canadian gourmet cookie brand built around culinary craftsmanship, experimental baking, and customer experience. The business combines a growing physical retail footprint with a digital-first omnichannel presence spanning Mobile App, Third-Party Delivery, and E-commerce.

Dataset Architecture

The analytics engine is a relational database containing 824,838+ transaction records and multiple interconnected operational tables covering retail transactions, orders, customers, products, stores, suppliers, and fulfillment.

Table	Role	Key Relationship / Purpose
transactions	Fact	High-level transactions, payment status, channel, revenue and refunds
order_details	Fact / Bridge	Order-level product quantities and line totals
customers	Dimension	Customer identity and geographic information

Table	Role	Key Relationship / Purpose
products	Dimension	Product catalog and categories
stores	Dimension	Physical store and location information
suppliers	Dimension	Supplier/vendor information
shipments	Logistics / Operational	Shipment, supplier and delivery status tracking

Core Data Relationships

- transactions.CustomerID → customers.CustomerID
- transactions.StoreID → stores.StoreID
- transactions.OrderID → order / order-detail processes
- order_details.ProductID → products.ProductID
- shipments.OrderID → order / transaction process
- shipments.SupplierID → suppliers.SupplierID

Analysis Scope & Reference Date

MODELING DECISION

The dataset contains 824,838+ transaction records. For revenue, RFM and customer analyses, Failed and Cancelled transactions are excluded wherever the SQL logic explicitly applies this filter. RFM and churn calculations use January 1, 2027 as the reference date for Recency and Days Since Last Order.

4. Data Dictionary — Analytical Fields

The following fields were central to the analytical workflow. The dictionary is intentionally focused on the fields used to explain business performance rather than listing every operational attribute.

Table	Field	Business Meaning
transactions	TransactionID	Unique transaction identifier
transactions	OrderID	Customer order identifier
transactions	CustomerID	Customer linked to the transaction
transactions	TransactionDate	Transaction timestamp
transactions	TransactionType	Purchase / refund type
transactions	PaymentMethod	Payment method used
transactions	Amount	Gross transaction value
transactions	RefundAmount	Amount refunded
transactions	PaymentStatus	Successful, cancelled, failed, refunded, etc.
transactions	StoreID	Store / channel origin identifier

Table	Field	Business Meaning
transactions	Channel	Sales channel
customers	CustomerID	Unique customer identifier
customers	CustomerName	Customer name
customers	City	Customer city
products	ProductID	Unique product identifier
products	ProductName	Product name
products	Category	Product category
order_details	OrderID	Order linked to product line
order_details	ProductID	Product linked to order line
order_details	Quantity	Units purchased
order_details	LineTotal	Line-level sales value
stores	StoreID	Unique store identifier
stores	StoreName	Store / branch name
stores	Location	Store geographic location
suppliers	SupplierID	Unique supplier identifier
shipments	ShipmentID	Unique shipment identifier
shipments	ShipmentDate	Shipment date
shipments	DeliveryDate	Delivery date
shipments	DeliveryStatus	Fulfillment status

5. Data Preparation & Analytical Controls

Before interpreting results, the analysis applies practical controls to ensure that revenue and customer metrics are based on relevant transactions and that measures are calculated at the correct business grain.

Key Controls Applied

- Failed and Cancelled transactions are excluded from revenue and customer analyses where specified in SQL.
- Revenue is separated into Gross Revenue, Refunds, and Net Revenue to avoid treating refunded value as retained sales.
- Customer frequency is based on distinct orders rather than raw transaction rows.
- RFM is calculated at customer level using Recency, Frequency, and Monetary behavior.
- Product performance is linked through order_details to preserve product-level sales detail.
- Refund analysis requires positive refund amounts before ranking refund hotspots.
- Churn-risk analysis uses days since the customer's latest purchase relative to the defined reference date.
- Cohort analysis maps customers to their first purchase month and tracks activity in later months.

ANALYST NOTE

The most important quality principle is metric consistency: the same business definition is used in SQL, Power BI measures, dashboard labels, and the written report.

6. Analytical Question Framework

The SQL phase was designed around business questions rather than isolated SQL features. The purpose was to turn the relational dataset into measurable answers that could later be communicated through Power BI.

Theme	Business Question	Primary Output
Revenue & Channel	Which channels and payment methods generate the strongest revenue, and how do refunds affect them?	Gross, Refunds, Net Revenue, Refund %, ATV
Sales Trend	How does revenue change over time?	Monthly revenue trend
Customer Intelligence	Which customer segments are most valuable?	RFM segments, spend, frequency
Product Performance	Which products and categories drive sales?	Orders, units, gross revenue
Store Performance	Which stores generate the strongest revenue?	Store ranking / regional comparison
Customer Value	Who are the highest-value customers?	Lifetime spend and order activity
Churn Risk	Which high-value customers may be at risk?	Days since last order and historical spend
Refund Hotspots	Which products create the greatest refund exposure?	Refund amount and refund rate
Retention	How well do customers remain active after acquisition?	Cohort activity by month

STORYTELLING APPROACH

The analysis moves from business performance → customer behavior → operational risk → retention, making the project easier for a decision-maker to follow than a collection of unrelated SQL queries.

7. SQL Server Analysis

This section presents the core SQL analysis behind the project. Each query is framed around a specific business question, followed by the key insight it produced and the resulting business implication.

7.1 Revenue, Channel & Transaction Performance

Business Question	Which sales channels generate the strongest commercial performance, and how much revenue is lost through refunds?
Analytical Approach	Transactions were aggregated by Channel and PaymentMethod. Gross Revenue, Total Refunds, Net Revenue, Refund Percentage, and Average Transaction Value were calculated after excluding Failed and Cancelled payments.
Result / Insight	In-Store is the leading channel by net revenue, with Mobile App and Website as meaningful secondary channels; Third-Party Delivery contributes the least. Credit

	Card is the dominant payment method at approximately 36M revenue.
Business Impact	The company should protect the strength of the In-Store channel while continuing to improve digital channels, and monitor refund performance alongside revenue so growth is not achieved at the expense of customer experience.

```

SELECT
  Channel,
  PaymentMethod,
  COUNT(TransactionID) AS TotalTransactions,
  SUM(Amount) AS GrossRevenue,
  SUM(RefundAmount) AS TotalRefunds,
  (SUM(Amount) - SUM(RefundAmount)) AS NetRevenue,
  CAST(SUM(RefundAmount) * 100.0 /
    NULLIF(SUM(Amount), 0) AS DECIMAL(5,2)) AS RefundPercentage,
  CAST(AVG(Amount) AS DECIMAL(10,2)) AS AverageTransactionValue
FROM transactions
WHERE PaymentStatus NOT IN ('Failed', 'Cancelled')
GROUP BY Channel, PaymentMethod
ORDER BY NetRevenue DESC;

```

7.2 Monthly Sales Trend & Growth

Business Question	How does sales performance change over time, and are there recurring seasonal patterns?
Analytical Approach	Transaction dates were aggregated into monthly periods to build a time-series view of revenue, later visualized in Power BI as a line chart for faster trend recognition.
Result / Insight	Monthly Net Revenue shows recurring peaks, especially around year-end periods, indicating a repeatable seasonal pattern.
Business Impact	A seasonal revenue pattern allows management to plan inventory, staffing, marketing, and promotional activity ahead of peak demand periods.

7.3 Customer RFM Segmentation

Business Question	Which customers are most valuable, and which customers require retention attention?
Analytical Approach	Recency measures days since the latest purchase, Frequency measures distinct orders, and Monetary measures total spend. Customers are scored using quintiles and grouped into four business segments.
Result / Insight	Champions/VIP customers demonstrate the strongest monetary value and purchasing frequency.
Business Impact	There is an opportunity to protect high-value relationships while using targeted campaigns to move Potential Loyalists upward and reactivate At Risk customers.


```

WITH RFM_Raw AS (
    SELECT
        CustomerID,
        DATEDIFF(DAY, MAX(CAST(TransactionDate AS DATE)), '2027-01-01') AS Recency,
        COUNT(DISTINCT OrderID) AS Frequency,
        SUM(Amount) AS Monetary
    FROM transactions
    WHERE PaymentStatus NOT IN ('Failed', 'Cancelled')
    GROUP BY CustomerID
),
RFM_Scores AS (
    SELECT
        CustomerID, Recency, Frequency, Monetary,
        NTILE(5) OVER (ORDER BY Recency DESC) AS R_Score,
        NTILE(5) OVER (ORDER BY Frequency ASC) AS F_Score,
        NTILE(5) OVER (ORDER BY Monetary ASC) AS M_Score
    FROM RFM_Raw
)
SELECT CustomerID, Recency, Frequency, Monetary,
       (R_Score + F_Score + M_Score) AS RFM_Total_Score
FROM RFM_Scores;

```

Customer Segments

Segment	RFM Rule	Business Meaning
Champions / VIP	RFM total ≥ 13	Highest-value, highly engaged customers
Loyal Customers	RFM total ≥ 10	Strong repeat customers with good value
Potential Loyalists	RFM total ≥ 7	Customers with potential to become more valuable
At Risk / Churned	RFM total < 7	Low engagement or declining customer activity

7.4 Product & Category Performance

Business Question	Which products drive the most orders, units, and gross revenue?
Analytical Approach	Order-line data was joined to the product catalog and aggregated by product and category to compare order frequency, unit volume, and gross revenue.
Result / Insight	Cookie Cake (3) is the top product by gross revenue at approximately 7.1M; Cakes is the highest-performing category at approximately 18.3M. Holiday Cookie Tin leads on units sold at approximately 233K units — revenue leadership and unit-volume leadership are not the same thing.
Business Impact	Both revenue and unit-volume measures should be monitored together so portfolio decisions aren't distorted by a single metric.

```

SELECT TOP 10
    p.ProductID, p.ProductName, p.ProductCategory,
    COUNT(od.OrderID) AS TotalTimesOrdered,

```

```
SUM(od.Quantity) AS TotalUnitsSold,  
SUM(od.LineTotal) AS GrossRevenue  
FROM products p  
JOIN order_details od ON p.ProductID = od.ProductID  
GROUP BY p.ProductID, p.ProductName, p.ProductCategory  
ORDER BY GrossRevenue DESC;
```

7.5 Store Performance & Customer Lifetime Value

Store-wise analysis compares gross revenue across physical locations to benchmark leading branches. The Cookie Lab London Main Street is the strongest store at approximately 4.8M gross revenue. Customer lifetime value is approximated through cumulative historical spend and order activity, ranking the highest-spending customers to support VIP treatment and retention planning.

```
SELECT TOP 20  
    c.CustomerID,  
    CONCAT(FirstName, ' ', LastName) AS CustomerName,  
    c.City,  
    COUNT(DISTINCT t.OrderID) AS TotalOrders,  
    SUM(t.Amount) AS LifetimeSpend,  
    CAST(MAX(t.TransactionDate) AS DATE) AS LastPurchaseDate  
FROM customers c  
JOIN transactions t ON c.CustomerID = t.CustomerID  
WHERE t.PaymentStatus NOT IN ('Failed', 'Cancelled')  
GROUP BY c.CustomerID, CONCAT(FirstName, ' ', LastName), c.City  
ORDER BY LifetimeSpend DESC;
```

BUSINESS INSIGHT

Store benchmarking can reveal operational practices worth replicating, while customer lifetime value provides a financial basis for prioritizing high-value relationships.

7.6 Refund Hotspots & Churn Risk

Refund Hotspots

Products were ranked by total refund amount and refund rate to identify potential quality, fulfillment, or customer-experience issues.

```
SELECT TOP 10  
    p.ProductID, p.ProductName, p.ProductCategory,  
    COUNT(t.TransactionID) AS TotalOrders,  
    SUM(t.RefundAmount) AS TotalRefundAmount,  
    CAST(SUM(t.RefundAmount) * 100.0 /  
        NULLIF(SUM(t.Amount), 0) AS DECIMAL(5,2)) AS RefundRatePercentage  
FROM products p  
JOIN order_details od ON p.ProductID = od.ProductID  
JOIN transactions t ON od.OrderID = t.OrderID  
GROUP BY p.ProductID, p.ProductName, p.ProductCategory  
HAVING SUM(t.RefundAmount) > 0
```

```
ORDER BY TotalRefundAmount DESC;
```

Holiday Cookie Tin has the highest refund amount at approximately 0.76M, making it a clear product-level hotspot for further investigation.

High-Value Customer Churn Risk

Customers who historically spent significant amounts but have not purchased recently are flagged. Customers with more than 90 days since their last order are surfaced for retention attention.

```
SELECT TOP 20
    c.CustomerID,
    CONCAT(FirstName, ' ', LastName) AS CustomerName,
    c.Email,
    MAX(t.TransactionDate) AS LastPurchaseDate,
    DATEDIFF(day, MAX(CAST(t.TransactionDate AS DATE)), '2027-01-01') AS
DaysSinceLastOrder,
    SUM(t.Amount) AS HistoricalSpend
FROM customers c
JOIN transactions t ON c.CustomerID = t.CustomerID
WHERE t.PaymentStatus NOT IN ('Failed', 'Cancelled')
GROUP BY c.CustomerID, CONCAT(FirstName, ' ', LastName), c.Email
HAVING DATEDIFF(day, MAX(CAST(t.TransactionDate AS DATE)), '2027-01-01') > 90
ORDER BY HistoricalSpend DESC;
```

7.7 Cohort & Customer Retention

Business Question	How effectively does the business retain customers after their first purchase?
Analytical Approach	Each customer is assigned to the month of their first purchase. Subsequent transactions are mapped back to that acquisition cohort to measure active customers over time.
Result / Insight	Cohort analysis reframes the question from 'how many customers do we have' to 'how well do we retain customers after acquisition.'
Business Impact	This view is critical for evaluating the quality of customer acquisition and the long-term value of marketing spend.

```
WITH FirstPurchase AS (
    SELECT CustomerID,
           MIN(FORMAT(CAST(TransactionDate AS DATE), 'yyyy-MM')) AS CohortMonth
    FROM transactions
    WHERE PaymentStatus NOT IN ('Failed', 'Cancelled')
    GROUP BY CustomerID
),
CustomerActivity AS (
    SELECT DISTINCT
        fp.CohortMonth,
        FORMAT(CAST(t.TransactionDate AS DATE), 'yyyy-MM') AS ActivityMonth,
        t.CustomerID
    FROM transactions t
    JOIN FirstPurchase fp ON t.CustomerID = fp.CustomerID
```

```
JOIN FirstPurchase fp ON t.CustomerID = fp.CustomerID
WHERE t.PaymentStatus NOT IN ('Failed', 'Cancelled')
)
SELECT CohortMonth, ActivityMonth, COUNT(DISTINCT CustomerID) AS ActiveCustomers
FROM CustomerActivity
GROUP BY CohortMonth, ActivityMonth
ORDER BY CohortMonth, ActivityMonth;
```

8. From SQL Analysis to Power BI

After completing the SQL analysis, the project moved from discovery to communication. Power BI was used to convert the analytical results into an interactive reporting layer that management can use to monitor performance on an ongoing basis.

SQL → POWER BI

SQL answers the business questions and creates the analytical foundation. Power BI turns those answers into a visual decision-support experience.

Dashboard Design Principles

- Start with executive KPIs before detailed visuals.
- Group visuals by business theme rather than by SQL query order.
- Use consistent definitions for Gross Revenue, Net Revenue, and Refunds.
- Use customer segmentation to move from descriptive reporting toward action.
- Use product and store rankings to identify performance leaders and hotspots.
- Keep the four-page structure focused, avoiding pages that don't change a business decision.

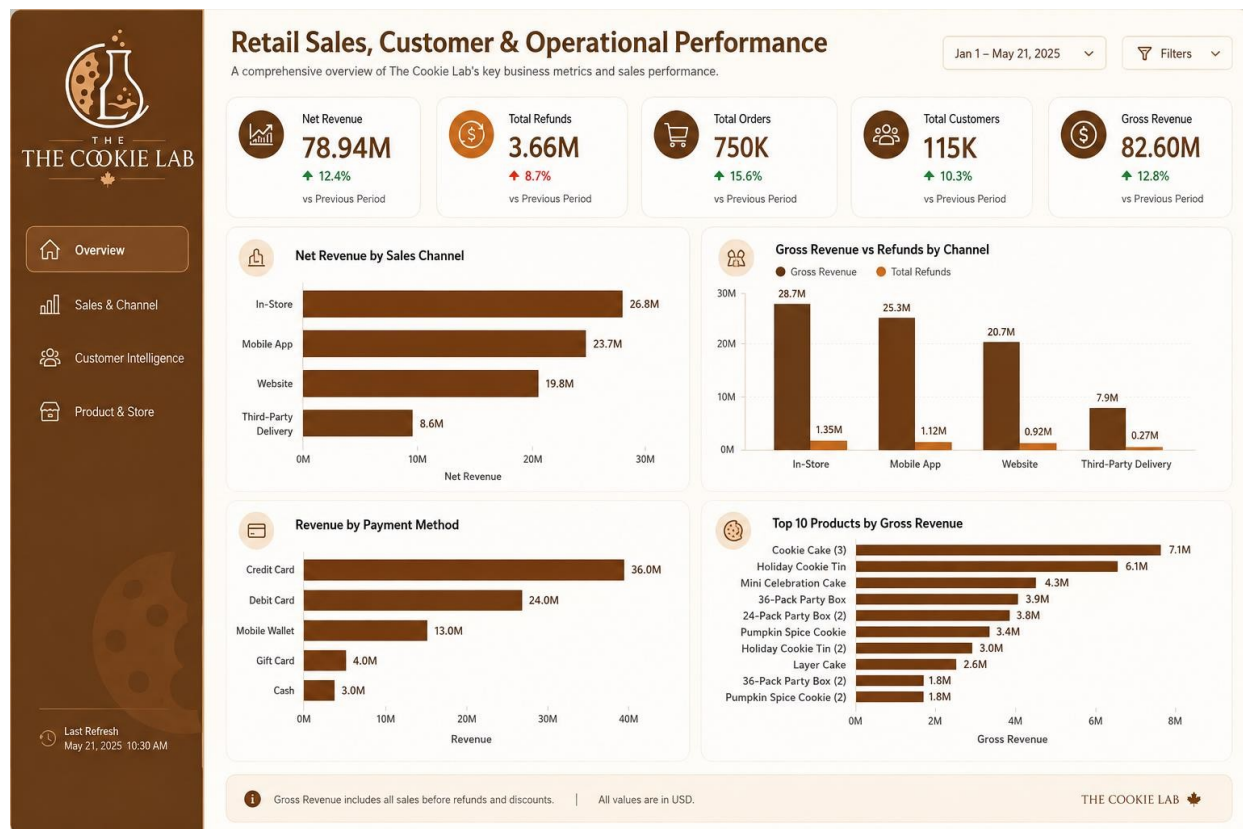
9. Power BI Dashboard Overview

The reporting layer is organized into four focused pages, each built around a specific decision theme.

Page	Title	Primary Decision Theme
1	Retail Sales, Customer & Operational Performance	Executive commercial overview
2	Sales & Channel Performance	Revenue trends, transaction behavior, refunds and stores
3	Customer Intelligence & RFM Analysis	Customer value, segmentation and retention
4	Product & Store Performance	Products, categories, units, refunds and stores

10. Power BI Page 1 — Retail Sales, Customer & Operational Performance

Page 1 acts as the executive overview. It brings the most important commercial KPIs and revenue drivers into a single screen so management can understand the overall business position before moving into detailed analysis.



What the Page Solves

- Provides Net Revenue, Gross Revenue, Total Refunds, Total Orders and Total Customers at a glance.
- Compares Net Revenue across sales channels.
- Compares Gross Revenue and Refunds by channel.
- Shows revenue contribution by payment method.
- Highlights the Top 10 Products by Gross Revenue.

Key Results

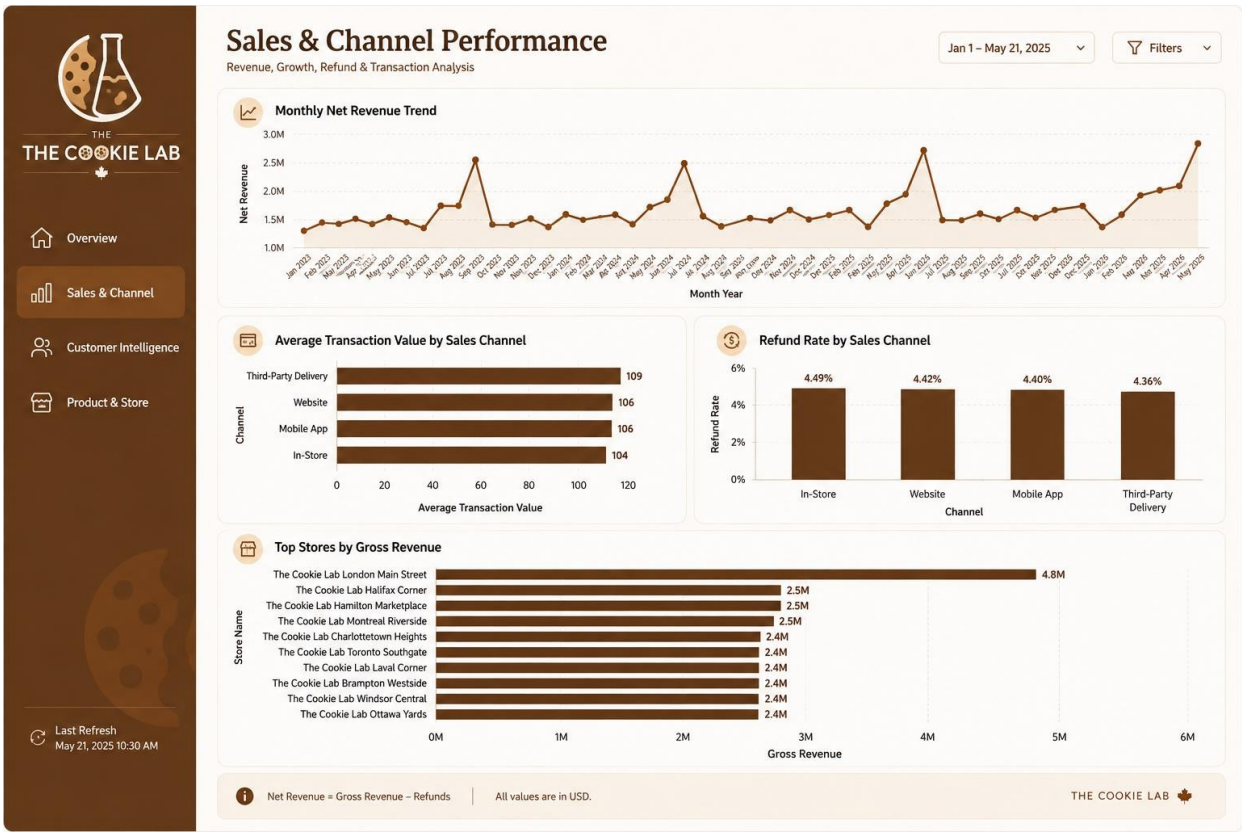
- 82.60M Gross Revenue and 78.94M Net Revenue.
- 3.66M Total Refunds.
- 750K Total Orders and 115K Total Customers.
- In-Store is the leading sales channel.
- Credit Card generates approximately 36M revenue.
- Cookie Cake (3) leads product gross revenue at approximately 7.1M.

Business Impact

This page provides a single executive snapshot of revenue, customer scale, channel performance, payment behavior and product performance. It answers the first management question: how is the business performing overall, and what are the major drivers?

11. Power BI Page 2 – Sales & Channel Performance

Page 2 moves from headline numbers into trend and channel behavior. It helps management understand not only how much the business sells, but when sales occur, how transaction value differs by channel, and where refund exposure appears.



Key Results

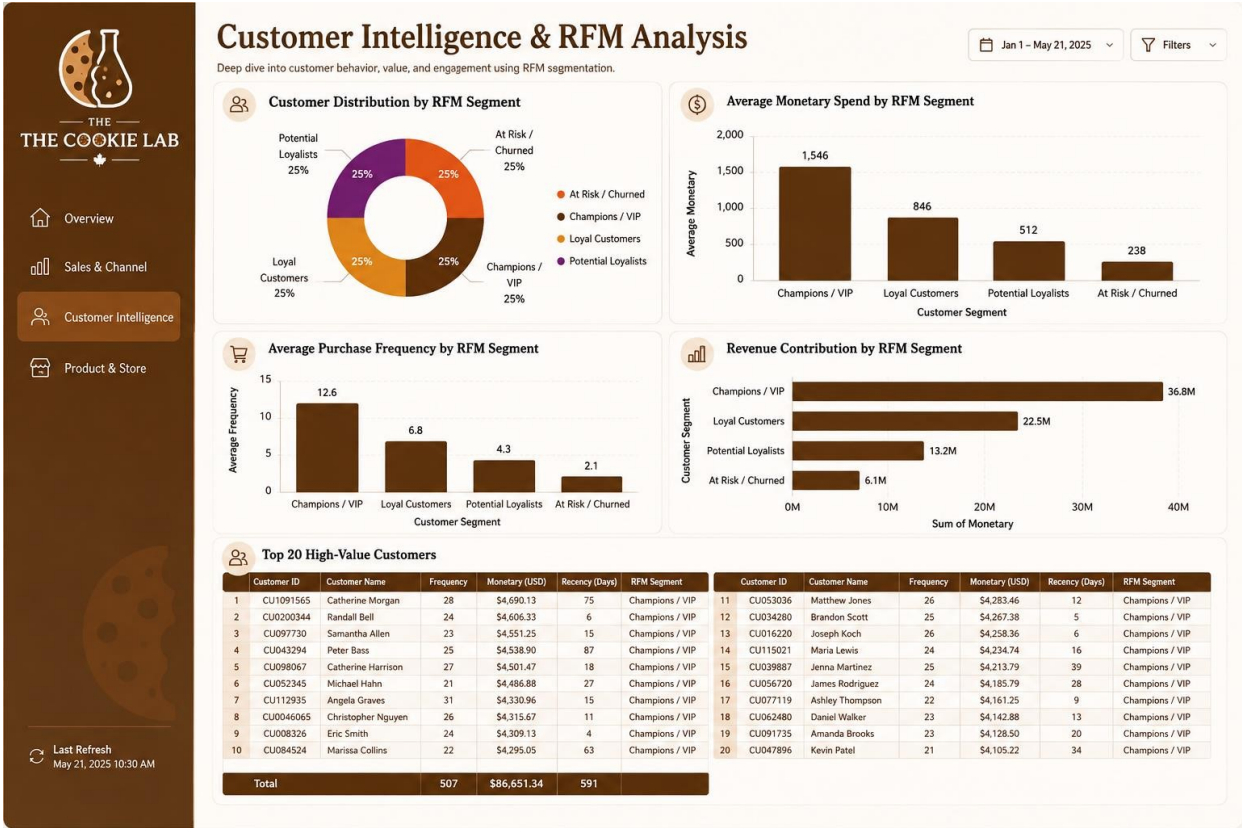
- Monthly Net Revenue shows a recurring seasonal pattern with stronger peaks around year-end periods.
- Third-Party Delivery has the highest Average Transaction Value at approximately 109.
- In-Store has the lowest Average Transaction Value among the four channels at approximately 104.
- Refund rates are relatively consistent, ranging from approximately 4.36% to 4.49%.
- In-Store has the highest refund rate at approximately 4.49%.
- The Cookie Lab London Main Street is the leading store at approximately 4.8M gross revenue.

Business Impact

The page supports sales planning, channel strategy, refund monitoring and store benchmarking. Seasonal patterns can also inform inventory, staffing and promotional planning.

12. Power BI Page 3 – Customer Intelligence & RFM Analysis

Page 3 focuses on the customer rather than the transaction. RFM segmentation converts purchase history into business segments that can be targeted differently.



Key Results

- Customers are grouped into Champions/VIP, Loyal Customers, Potential Loyalists, and At Risk/Churned.
- Champions/VIP customers demonstrate the strongest monetary value and purchasing frequency.
- Revenue contribution is concentrated more heavily in high-value customer segments.
- The Top 20 High-Value Customers table identifies customers with the strongest spending and purchase activity.

Business Impact

The page enables differentiated customer strategies: protect VIP relationships, nurture Potential Loyalists, and reactivate At Risk/Churned customers. This moves the project from descriptive reporting toward customer retention action.

13. Power BI Page 4 – Product & Store Performance

Page 4 brings together product sales, category performance, unit volume, refund exposure and store performance. It is designed to help commercial and operational teams identify what sells, what moves in volume, and where risk exists.



Key Results

- Cookie Cake (3) is the top product by gross revenue at approximately 7.1M.
- Cakes is the highest-performing product category at approximately 18.3M revenue.
- Holiday Cookie Tin leads units sold at approximately 233K units.
- Holiday Cookie Tin also has the highest refund amount at approximately 0.76M.
- The Cookie Lab London Main Street is the top store at approximately 4.8M gross revenue.

Business Impact

This page supports product portfolio decisions, inventory planning, quality investigation and store benchmarking. Importantly, it separates revenue leadership from volume leadership so management does not rely on a single KPI.

14. Cross-Dashboard Business Insights

Viewed together, the four Power BI pages tell a consistent business story.

- Overall commercial performance is strong, but refunds create a measurable gap between gross and retained revenue.
- In-Store remains the strongest commercial channel, while digital channels represent meaningful secondary growth opportunities.
- Revenue follows a recurring seasonal pattern, making advance planning important ahead of peak periods.
- A relatively small group of high-value customers contributes disproportionately to monetary performance, justifying targeted retention strategies.

- Customer segmentation creates concrete opportunities for VIP retention and win-back campaigns aimed at At Risk customers.
- Product performance differs depending on whether revenue or unit volume is the measure — the top revenue product is not necessarily the top volume product.
- Holiday Cookie Tin stands out as a significant refund hotspot that warrants product-level investigation.
- Leading stores such as London Main Street provide operational benchmarks that other locations can be measured against.
- Together, these findings translate into specific, prioritized management actions rather than a list of disconnected observations.

THE BUSINESS STORY IN ONE LINE

The Cookie Lab has strong commercial performance, but the next level of growth depends on protecting high-value customers, improving digital channels, controlling refund hotspots, and replicating best-performing product and store practices.

15. Business Recommendations

Strengthen High-Performing Channels

Why It Matters	In-Store drives the largest share of net revenue, and digital channels represent room to grow.
Suggested Action	Protect In-Store performance while developing Mobile App and Website through targeted promotions and stronger digital customer experiences.
KPI to Monitor	Net Revenue by Channel, Average Transaction Value, Channel Growth

Investigate Refund Hotspots

Why It Matters	Holiday Cookie Tin carries the highest refund amount, signaling a possible quality or fulfillment issue.
Suggested Action	Prioritize products with high refund amounts and refund rates — particularly Holiday Cookie Tin — for root-cause investigation.
KPI to Monitor	Refund Amount, Refund Rate, Product-level complaints/returns

Prioritize High-Value Customers

Why It Matters	Champions/VIP customers show the strongest monetary value and purchase frequency.
Suggested Action	Use RFM segmentation to create VIP loyalty benefits for Champions/VIP and personalized retention offers for Loyal Customers.
KPI to Monitor	Repeat Purchase Rate, Monetary Value, RFM Segment Migration

Reactivate At-Risk Customers

Why It Matters	High-spend customers with deteriorating recency represent recoverable revenue.
Suggested Action	Create targeted win-back campaigns for high-spend customers whose purchase recency has declined.
KPI to Monitor	Reactivation Rate, Days Since Last Order, Recovered Revenue

Optimize the Product Portfolio

Why It Matters	Revenue leadership and unit-volume leadership are not the same products.
Suggested Action	Promote high-revenue products while separately monitoring high-volume products to understand the difference between units and value.
KPI to Monitor	Gross Revenue, Units Sold, Revenue per Unit, Refund Rate

Benchmark Store Performance

Why It Matters	London Main Street significantly outperforms other locations.
Suggested Action	Use leading stores as benchmarks and investigate the operational factors behind their stronger revenue.
KPI to Monitor	Store Revenue, Average Order Value, Growth, Refund Rate

Establish Continuous BI Monitoring

Why It Matters	A one-time report loses value once conditions change.
Suggested Action	Use the Power BI dashboard as an ongoing management tool rather than a one-time analysis.
KPI to Monitor	Revenue, Refunds, Customer Segments, Product and Store KPIs

16. Final Conclusion

The Cookie Lab project demonstrates an end-to-end Data Analyst workflow: understanding the business, working with a relational dataset, translating business questions into SQL, identifying meaningful patterns, and communicating those findings through an interactive Power BI dashboard.

The analysis reflects a strong commercial position — 82.60M gross revenue, 78.94M net revenue, 750K orders and 115K customers — alongside 3.66M in refunds that represents a clear, addressable area for operational improvement.

The most valuable outcome of this project is not any single KPI. It is the connection between revenue performance, customer value, product behavior, store performance, and operational risk. SQL provided the analytical depth; Power BI turned that depth into a decision-support experience that a business can act on and continue to monitor.

17. Analyst Perspective

This project reflects an analytical approach grounded in business relevance rather than technical demonstration for its own sake. Every query, segment, and visual was built to answer a specific question a retail business would actually ask.

It demonstrates the ability to frame a business problem, build multi-table SQL logic — including joins, CTEs, window functions, and aggregation — and translate the results into customer, product, and store-level insight. It also demonstrates the discipline of connecting each finding to a recommended action and a KPI to monitor, rather than stopping at description.

GUIDING PRINCIPLE

A strong analytics project does not stop at 'what happened.' It explains why the result matters, who should act, what should be monitored next, and how success can be measured.

18. Technical Skills & Tools

Core Tools

Tool	Purpose
Microsoft SQL Server	Relational data storage and analytical query engine
SQL Server Management Studio (SSMS)	Query development and analysis environment
Power BI	Dashboard design, data visualization and business monitoring

Analytical Techniques Demonstrated

- SQL aggregation and multi-table JOINS
- Common Table Expressions (CTEs)
- Window functions (NTILE)
- RFM segmentation
- Customer lifetime value analysis
- Cohort and retention analysis
- Monthly trend analysis
- Refund and churn-risk analysis
- Product and category performance analysis
- Store benchmarking
- KPI design for dashboards

19. Portfolio Project Summary

This case study was prepared as an independent Data Analyst portfolio project using The Cookie Lab as the business case context. It demonstrates a complete workflow from a relational SQL Server dataset through business-question-driven analysis to a four-page Power BI reporting solution, culminating in decision-ready recommendations.

Item	Detail
Project Type	Data Analyst Portfolio Case Study
Business Context	The Cookie Lab — Canadian premium omnichannel cookie retailer
Prepared By	Khurram Naveed, Data Analyst
Core Tools	Microsoft SQL Server (SSMS), Power BI
Data Scale	824,838+ transaction records
Reporting Layer	Four-page Power BI dashboard